

ICTs For 21st Century Learning



Atul M. Kulkarni is an M.Tech in Communication Engineering from National Institute of Technology, Karnataka. Currently, he is working as a lecturer at Dr Babasaheb Ambedkar Technological University, Lonere, Raigad

Information and communication technologies (ICTs) have brought about a paradigm shift in education. Online educational methodology and tools like massive open online course (Mooc), modular object-oriented dynamic learning environment (Moodle), videoconferencing, virtual laboratories, screencasts, e-mail, forums, search engines, blogs, podcasts, vodcasts, wikis, vokus, high-tech 3D graphics, e-Groups, learning dialogue videos, learning-by-doing videos, digital game/toy-based learning and flipped-classrooms with associated pedagogy of active learning have quickly emerged as the backbone of effective education in this century. In fact, blended learning is the necessity of 21st century.

Benefits

ICTs provide a great solution for information collection and sharing, making learning more relaxed and productive—that too for all, anytime and anywhere. Tools like free and open source software (FOSS), and websites like github.com and SourceForge.net, help students with knowledge building, reflection, sharing and collaboration in project work. ICTs also promote self-evaluation and self-study via online tools. Users can learn in their own dialects, which fills the gap of language barrier in teaching-learning process.

Integration of ICTs in education helps students, teachers and administrators

alike. ICTs help where traditional methods of learning may not be effective, such as for students with learning disabilities like dyscalculia and dyslexia. Research and revision can be done on the Internet as well as by using storage media like CD-ROMs and USB drives and application programs. In fact, application programs with animation, video, quizzes and sound make learning more interactive.

Teachers and administrators use ICT tools like electronic data interchange or management information system for quick and accurate exam entries or other data logging operations. This reduces expenditure (printing and posting) and errors as well. As there is less paper work, it's more environment-friendly and saves teachers' time. Attendance monitoring software can automatically send a message to parents if their child is absent from class. Some schools are now using ICT to manage their day-to-day activities and monitor students. Innovative teaching-learning practices and methodologies are emerging with integration of ICT in learning.

Tools and methodologies employed

Digital games and toys. Digital games and toys shape thinking and creativity ability of learners. Animated cartoon learning videos, storytelling toys, and educational learning tablets and computer-like gadgets have emerged as guided and effective learning tools for kids. Well-designed collaborative 3D learning games are emerging as a future learning tool for smart class-

Computer-like gadgets have emerged as guided and effective learning tools for kids (Image courtesy: www.amazon.in)



TABLE I
TRADITIONAL VS ICT-BASED EDUCATIONAL APPROACH

Traditional approach	ICT based approach
Teacher acts as a sage on the stage	Teacher acts as a guide on the side
Classroom is limited by its physical size	Teaching a large number of students is possible
Content delivery is synchronous	Content delivery is anytime, anywhere, not synchronous
Content delivered using PowerPoint presentations, blackboards, textbooks, videos, etc	Content delivered via multimedia, simulation, video-on-demand, videoconferencing, digital library, etc
Teacher-centered classroom	Student-centered classroom